Rohit Verma

Bachelor of Technology
in Computer Science and Engineering
Indian Institute of Technology, Ropar

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Education

Degree	Institute/Board	CGPA/Percentage	Year
Bachelor of Technology	Indian Institute of Technology, Ropar	7.55 (Till 6th Sem)	2023-2027
Senior Secondary	Central Board of Secondary Education	88.2%	2023
Secondary	Central Board of Secondary Education	94.6%	2021

Experience

- **AlgoUniversity** *Jun 2026 - July 2026*
  Software Engineer Intern
  – Engineered AI-assisted frontend templates for small-business clients using Context7 MCP and Ralph-loop agent workflows.
  – Automated deployment with a GitHub Actions + Vercel CI/CD pipeline, streamlining build and release cycles.
  – Developed backend and frontend features for a full-stack module management platform, including CRUD and integration workflows.
  – Mentored 15+ students weekly on AI-assisted frontend development.

Projects

Snehita Well-Being (Mental Health Counselling Platform, IIT Ropar) *Feb 2026 – May 2026*
Full-Stack Web Application Github
- Built a secure full-stack platform for mental health appointment scheduling and management.
- Developed RESTful APIs with JWT authentication and role-based access control.
- Designed a responsive React frontend for booking, profiles, and availability management.
- Architected an MVC backend with PostgreSQL and containerized deployment using Docker.
– **Tech Stack:** React.js, Node.js, Express.js, PostgreSQL, Docker, JWT

AIMS – Academic Information Management System *Jan 2026 – Feb 2026*
Full-Stack Role-Based Web Application Github
- Developed a role-based academic management system supporting Student, Instructor, Advisor, and Admin workflows.
- Implemented multi-level course enrollment approval (Instructor → Faculty Advisor) reflecting real university processes.
- Designed OTP-based email authentication with role-based access control and auto dashboard redirection.
- Built RESTful APIs with Express.js and integrated Supabase (PostgreSQL) using UUID keys and relational constraints.
– **Tech Stack:** React, Node.js, Express.js, Supabase (PostgreSQL), UUID, Nodemailer, REST APIs

Rank Aggregation with Proportionate Fairness (RAPF) *Sept 2025 – Nov 2025*
Database Management Systems Github
- Implemented a fair rank aggregation system ensuring proportionate representation of protected groups using p-fairness.
- Engineered an exact greedy algorithm ($O(n)$) and graph-matching approximation to solve Individual p-Fairness (IPF).
- Developed a deterministic framework using multiprocessing and Gurobi ILP, achieving 100% fairness vs 78% in baselines.
– **Tech Stack:** Python, Gurobi Optimizer, Multiprocessing, Graph Algorithms

RISC-V Assembler & Simulator *Mar 2025 – Apr 2025*
Computer Architecture Github
- Built an assembler to translate RISC-V assembly code into machine instructions with full ISA and directive support.
- Developed a 5-stage pipelined CPU simulator with configurable data forwarding and hazard resolution.
- Implemented branch prediction using BTB and a 1-bit predictor to reduce control-flow stalls.
– **Tech Stack:** C++, RISC-V ISA, Pipelining, Binary Encoding, Branch Prediction

Technical Skills

- **Programming Languages**: C/C++, Python, JavaScript
- **Development Tools & Frameworks**: HTML, CSS, React, Node.js, Express.js, PostgreSQL, Docker
- **DevOps & AI Tooling**: GitHub Actions, Vercel, MCP (Context7), AI-assisted agent workflows (Ralph loop)
- **Misc**: Git, GitHub, Vim, MySQL
- **Key Concepts**: Data Structures and Algorithms, OOPs, REST APIs, Computer Networks, Operating System

Key courses taken

- **CSE**: Computer Networks, Software Engineering, Operating System, Database Management System, Data Structure & Algorithm, Computer Architecture, Programming Paradigms and Pragmatics
- **Others**: Professional Communication, Financial Risk Management, Signal and System, Economics, Tinkering

Achievements

- **Codeforces:** Achieved a peak rating of 1210 and ranked in the top 10% in three Codeforces contests; view profile.
- **Problem Solving:** Practiced and solved 900+ problems across Codeforces, LeetCode, and GFG, strengthening core DSA concepts.